

Olevel economics

Chp:- Demand and Supply

Topic :- Indirect tax and PED!

Zarrak
Nushtay

Indirect tax and Price Elasticity of demand

e.g. sales tax



Cost of production ↑



Burden ↑



Supply ↓



Burden passed on to consumers



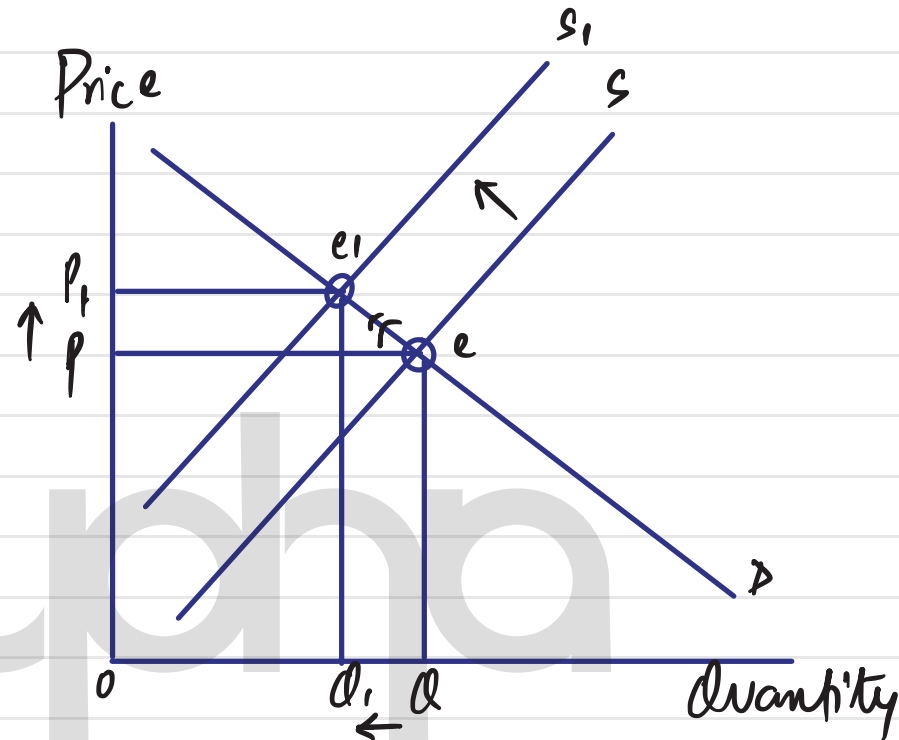
Price ↑



Quantity demanded ↓



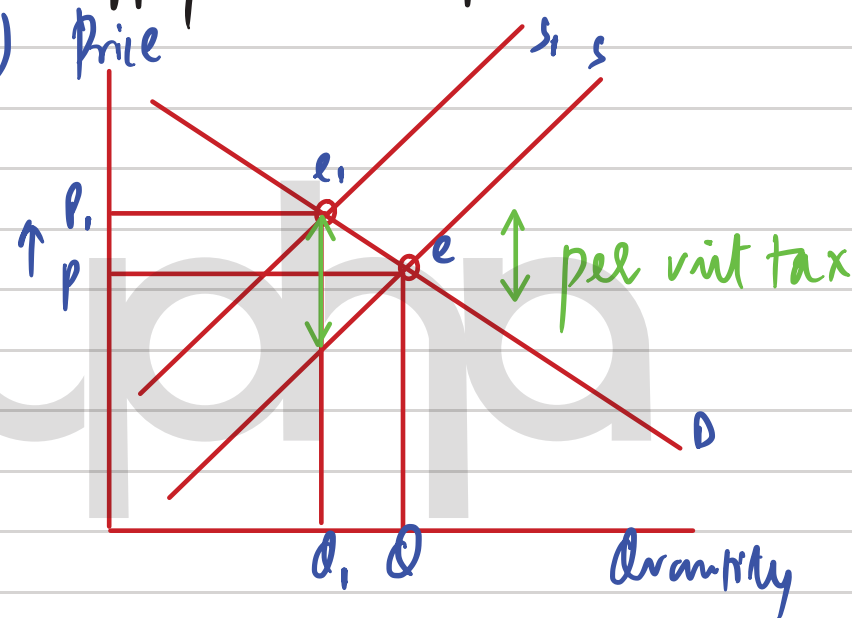
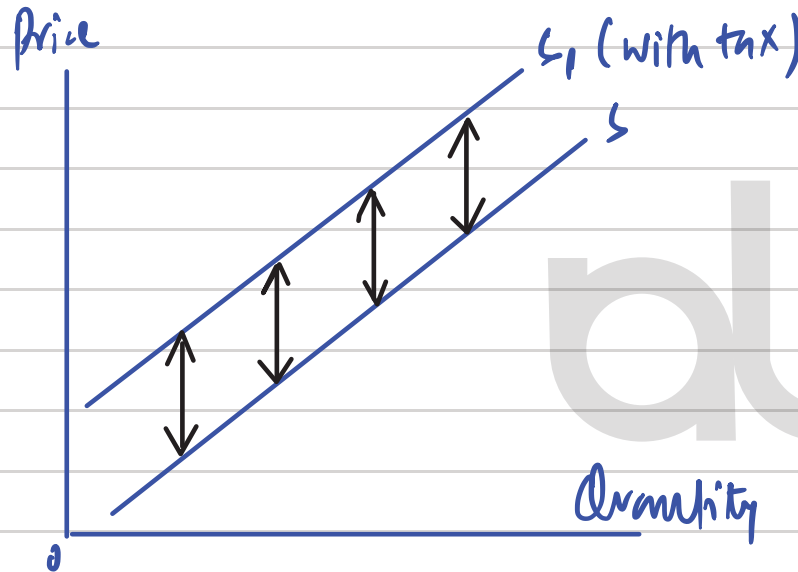
To what extent? It depends on PED!



Note:- Indirect tax, a tax which is imposed on goods and services, this increases its cost of production causing a reduction in supply, the burden is passed on to consumers in form of higher prices causing contraction in quantity demanded, the extent to which it contract depends on PED!

Concept of per unit tax, e.g. tax on bottle '\$1'

Indirect tax, if per unit, or specific tax, shifts supply curve parallel, therefore the vertical distance between two supply curve is per unit tax.



↑ This distance is per unit tax ↓

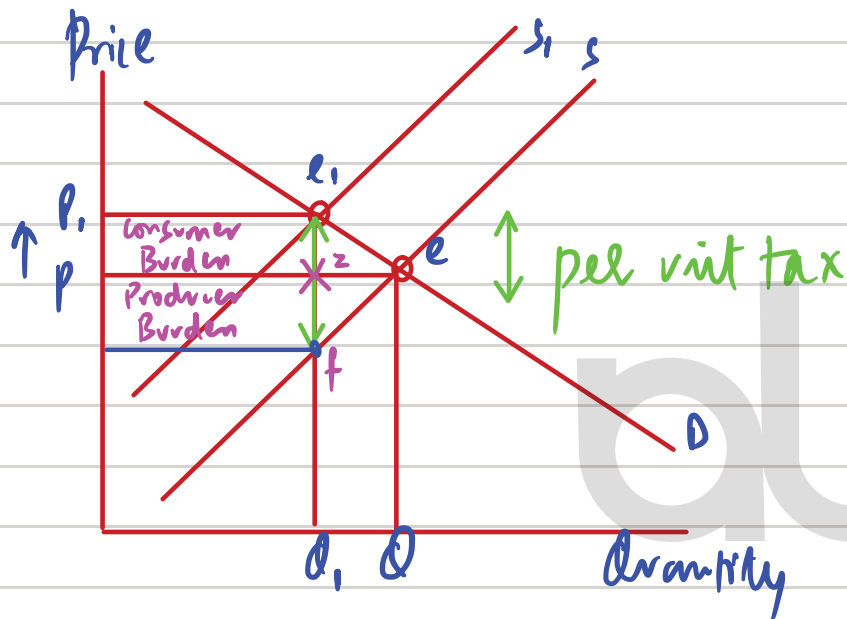
Once you have drawn the diagram, you need to find the vertical distance from new quantity traded that is Q_1 .

Final step is to divide the per unit tax between consumers (consumer burden, C.B) and producers (producer burden, P.B)

Now you need to divide the distance from initial equilibrium price

$$e_1 - f \text{ :- per unit tax}$$

From initial equilibrium price to new equilibrium price, this is represented as consumer burden, since consumers are paying more than before, that is, $e_1 - z$.



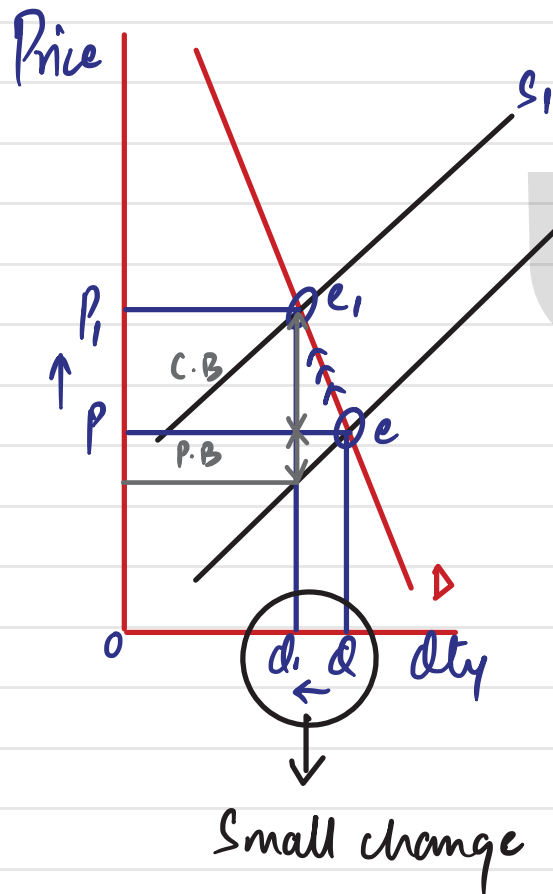
Importantly, there is still a distance left from per unit tax, that is not passed by producers onto consumers in form of higher prices, that is, $z - f$, this area is producer burden

Indirect tax → Always link to Government objectives e.g.



Imposed on a product with Inelastic demand

- 1) Tax revenue
- 2) Consumption ↓
- 3) Pollution ↓



* $C.B > P.B$

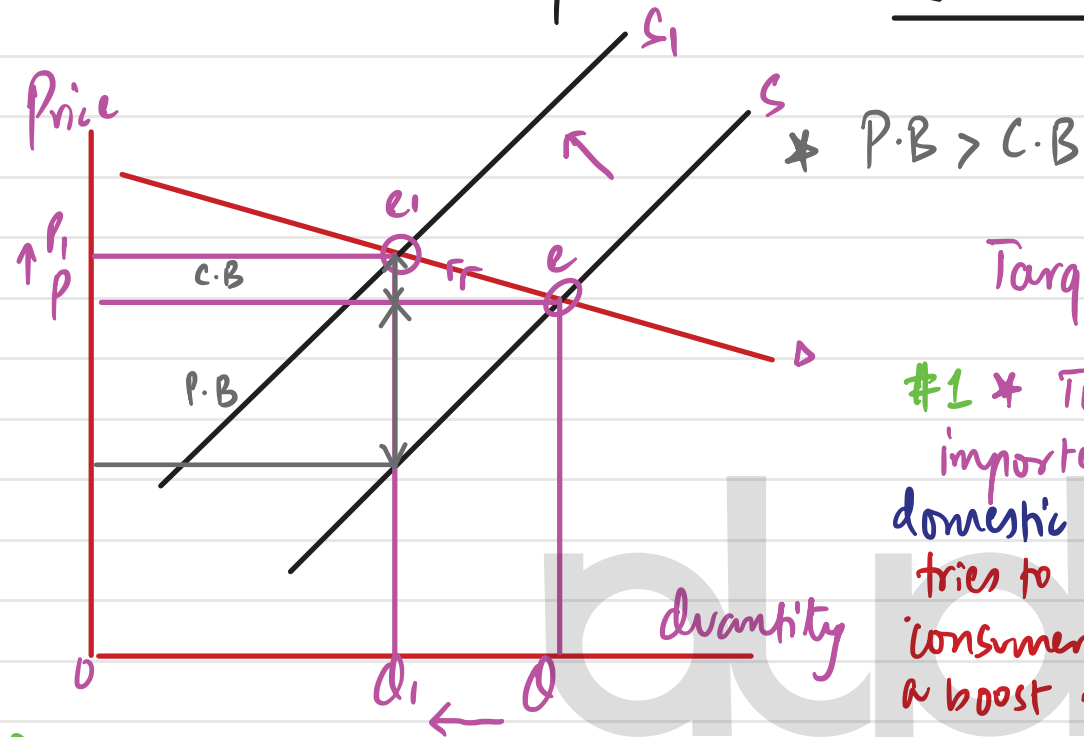
Points

A larger change in price brings a smaller percentage change in quantity demanded.

- ✧ Government is aware that demand for harmful goods is inelastic e.g. drugs
- ✧ It is difficult to reduce consumption but tax revenue can be generated
- ✧ Once tax is imposed, prices will increase by greater percentage but quantity will contract by smaller percentage, generating higher tax revenue which can be used by government on health!

Indirect tax on a product with elastic demand

⇒ A smaller change in price brings a larger percentage change in quantity demanded.



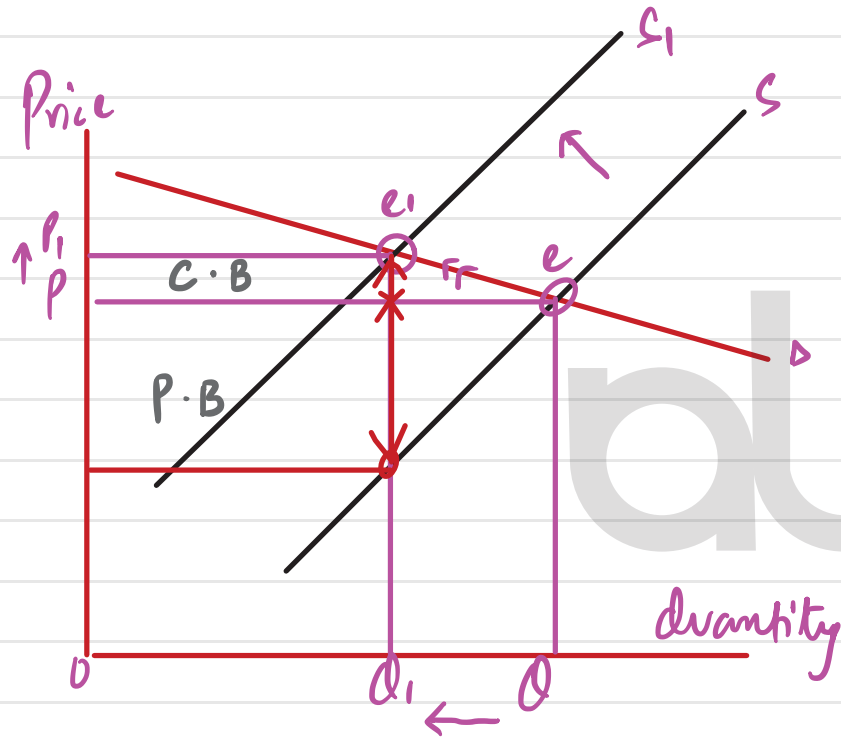
Target:-

#1 * To reduce the consumption of imported goods, as options are available in domestic market for e.g. cars. Government tries to raise the tax on imported cars so that consumers can switch to domestic cars, causing a boost to our economy.

#2

* Government is aware that use of cars can create pollution, so by imposing tax on cars, consumers might switch to public transport, causing a reduction in pollution and improvement of health in the economy!

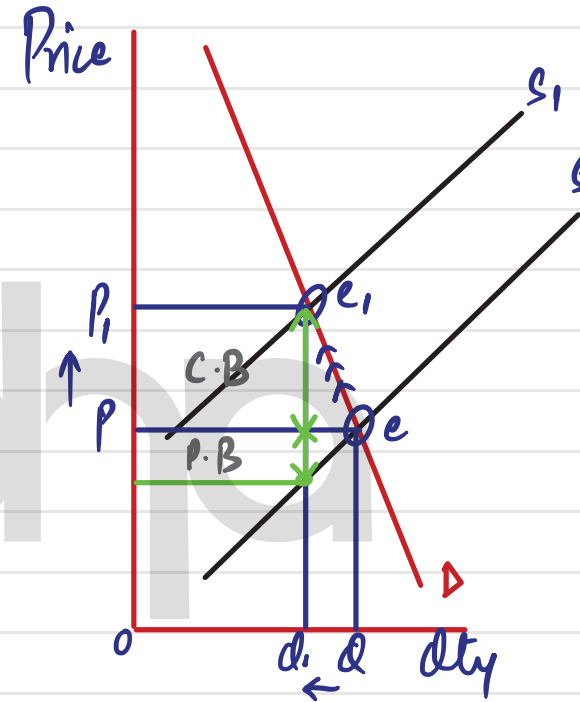
Identify per unit tax and C.B and P.B



C.B :- Consumer Burden
P.B :- Producer Burden

$P.B > C.B$

Elastic demand



$C.B > P.B$

Inelastic demand

Questions

Marks: 10

Q1 Using diagrams, explain how government can achieve its objectives with the knowledge of PED, by imposing indirect tax? 10

Ans:- * State the objectives (Tax revenue, consumption ↓, pollution ↓)

* Define indirect tax

* Impact of indirect tax on elastic demand (e.g. + Diagram)

* Impact of indirect tax on inelastic demand (e.g. + Diagram)

* 2 diagrams